Supervised Contrastive Learning-Based Unsupervised Domain Adaptation for Hyperspectral Image Classification

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***Abstract*— Deep domain adaptation has achieved promising results in cross-domain hyperspectral image (HSI) classification. However, existing methods often focus on aligning data distributions without sufficient consideration of separability of source and target domain data themselves. In addition, current adversarial domain adaptation methods aim to achieve similar distributions between domains by confusing the discriminator, rather than obtaining a more compact distribution. In particular, existing methods are not discriminative enough for the target domain due to the difficulty of obtaining high-confidence labeled samples of the target domain. To address the above challenges, we propose a supervised contrastive learning (SCL)-based unsupervised domain adaptation for HSI classification. An SCL strategy is then performed in both the source and target domains, which allows samples from the same category to be pulled closer together and samples from different categories to be pushed further apart, thus enhancing the separability of the data within the domain. The domain adaptation task is treated as a one-class classification (OCC) task, and a novel domain similarity loss based on OCC is introduced to reduce the discrepancy between domains. Finally, a confidence learning-based sample selection strategy is designed to select high-confidence labeled samples from the target domain to fine-tune the domain adaptation model, which can enhance the discrimination of the model to the target domain. Experimental results on three cross-domain datasets demonstrate that our proposed method outperforms existing domain adaptation methods. Our source code is available at https://github.com/Li-ZK/SCLUDA-2023.**

***Index Terms*— Classification, domain adaptation, hyperspectral image (HSI), supervised contrastive learning (SCL).**

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I. Introduction

HYPERSPECTRAL imaging is a powerful technique that has gained considerable attention in the field of remote sensing (RS). It captures a vast amount of spectral data across the electromagnetic spectrum, providing highly detailed images of the surface of the Earth. In the past few years, the classification of hyperspectral image (HSI) has gained significant attention as a popular research topic [1], [2], and it has found extensive applications in fields such as environmental and disaster monitoring [3], smart agriculture [4], [5], and military operations. HSI classification aims to assign a specific land-cover class to each pixel based on its spectral characteristics, thereby enabling accurate representation of land-cover distribution in the image under study [6].

Over the past few decades, there has been extensive research on various supervised classification algorithms, including random forest [7], extreme learning machines [8], and neural networks [9]. These conventional algorithms have been extensively studied with the aim of achieving accurate classification. However, the high cost and time-consuming nature of labeling RS data is widely recognized [10]. The insufficient availability of labeled samples presents a significant obstacle for accurate classification of HSIs [6]. Moreover, there can be spectral shifts [2] between HSIs that are captured at different times, locations, or by different sensors. Consequently, traditional classification models that are trained on one HSI may not achieve satisfactory performance on another image that exhibits spectral shift. Fortunately, domain adaptation [11], [12] can address this issue by transferring knowledge from a source HSI to improve classification accuracy on a target HSI with spectral shift. In this scenario, the HSI that has sufficient labeled samples is referred to as the source domain, while the HSI that has fewer labeled samples or no labeled samples is known as the target domain. Domain adaptation approaches can generally be divided into three main categories: instance-based approaches, classifier-based approaches, and feature-based approaches.

Instance-based approaches aim to reduce the discrepancy between the source and target domains by adjusting the weights of data instances. Huang et al. [13] proposed the kernel mean matching (KMM) method to reduce distribution mismatch between source and target domains. It does

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this by aligning the means of source and target samples in feature space through directly reweighting the source samples. Sugiyama et al. [14] introduced a method called Kullback–Leibler importance estimation procedure (KLIEP), which relies on minimizing the Kullback–Leibler divergence and includes an automatic model selection process. Yao and Doretto [15] proposed an algorithm called multisource TrAdaBoost (MsTrAdaBoost), which involves two main steps in each iteration. First, the MsTrAdaBoost algorithm trains a set of candidate weak classifiers using weighted instances from pairs of each source and target domain. Second, a classifier with the lowest classification error rate on target domain instances is selected and used to update the weights of instances in both the source and target domains. The fundamental assumption of this type of method is that the source and target domain data have similar or identical support, which means that they share some common features. However, in many practical scenarios, features of source and target domain data may not be exactly identical or overlapping, so the instance-based method cannot effectively reduce the discrepancy between domains.

The objective of classifier-based approaches is to train a classifier that can be adapted to the target domain, by utilizing labeled samples from the source domain. Joachims et al. [16] employed iterative training of a support vector machine (SVM) using both the source and target datasets, where the pseudo-labels assigned by the trained classifier were utilized in the process. Zhong and Zhang [17] proposed an iterative training sample update (ITSU) method based on posteriori spatial features. To be more specific, the ITSU approach involves iteratively creating training samples from the target domain, using spatially filtered posteriori probabilities with a feedback structure. At the same time, the samples used for training from the source domain are deleted. Ultimately, the ITSU approach produces a classifier that relies exclusively on the training samples obtained from the target domain. It is evident that assigning pseudo-labels to the target domain data is a highly effective method for cross-domain classification [18]. However, acquiring high-quality labels for the target domain poses a challenge, and assessing the accuracy of pseudo-labeling is problematic because there may be no available labels in the target domain.

Feature-based approaches rely on extracting common feature representations through either feature transformation [19], [20] or feature reconstruction [21], [22]. This allows the algorithm to adapt to the target domain by leveraging the similarities between the domains at the feature level. In recent years, deep neural networks have gained attention in the field of domain adaptation due to their excellent feature representation capabilities. To enhance the effectiveness of feature-based domain adaptation strategies, a common technique is to use a deep transfer network in conjunction with them. The deep domain adaptation approaches can be broadly categorized into two types: discrepancy-based approaches and adversarial-based approaches.

For discrepancy-based approaches, Long et al. [23] introduced a deep adaptation network (DAN) that extends deep convolutional neural networks (CNNs) to domain adaptation scenarios. The DAN approach employs the multiple-kernel maximum mean discrepancy (MK-MMD) metric to quantify the distribution discrepancy between domains. Sun and Saenko [24] proposed a technique called domain adaptation with correlation alignment (D-CORAL), which aligns the covariance matrices of the features obtained from deep networks in the two domains to make the learned features more domain-invariant. Kang et al. [25] introduced a novel objective function called contrastive domain discrepancy (CDD), which enables class-aware unsupervised domain adaptation (UDA). Wang et al. [26] introduced a technique for adapting to different domains in HSI classification using deep neural networks, which involves learning both discriminative features and manifold embedding to improve the performance.

For adversarial-based approaches, Ganin et al. [27] designed a domain adversarial neural network (DANN) to align feature distributions in two domains, where alignment is achieved through standard backpropagation training. Liu et al. [28] proposed a technique called classwise distribution adaptation (CDA) that creates domain-invariant features through the use of both classwise maximum mean discrepancy (MMD) and classwise adversarial adaptation. In addition, they incorporated pseudo-labels from target data during model optimization. Deng et al. [29] introduced the Euclidean distance-based deep metric model with UDA. This method not only utilizes an adversarial approach to ensure that the embedding features of both the source and target samples are indistinguishable but also encourages the formation of similar clusters between the embeddings of the target scene and the source scene. Fang et al. [30] and Huang et al. [31] employed a domain adaptation method based on biclassifiers, which replaces the domain discriminator with two different classifiers.

Although the deep domain adaptation approaches have been very successful, there are still some problems. First, current deep domain adaptation methods for HSI classification focus on aligning data distribution but rarely consider the separability of data itself. Second, the adversarial-based domain adaptation methods confuse the domain discriminator by adding a gradient reversal layer (GRL) [27], which achieves similar distributions between domains without stricter constraints. Moreover, due to the difficulty in obtaining high-confidence labeled samples of the target domain, existing deep domain adaptation methods are not discriminative enough for the target domain.

To address the above challenges, we propose a supervised contrastive learning-based UDA (SCLUDA) method for HSI classification. To enhance the separability of the data itself in the source and target domains, we extend the self-supervised contrastive learning to the fully supervised setting. In contrast to self-supervised contrastive learning [32], which uses only a single positive sample pair, supervised contrastive learning (SCL) [33] considers many positive sample pairs for each anchor in addition to many negative ones. In SCLUDA, the SCL is performed simultaneously in the source and target domains. We usually think that samples of the same class should be close together and samples of different classes should be far apart. By performing the SCL in both domains, data separability can be expanded. Since labels are not

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available for the target domain, the predicted labels of the target data are pseudo-labels to generate data pairs.

To achieve a more similar distribution between domains, the domain adaptation task is considered as a one-class classification (OCC) task [34], where a novel domain similarity loss (DSL) based on OCC is designed to project images from the source and target domains into a unified new nominal domain. Unlike current adversarial-based domain adaptation methods, the DSL based on OCC imposes stronger constraints, which makes the distribution closer. In addition, a local MMD (LMMD) strategy [35] is introduced to align the class-conditional distribution by considering different sample weights. Since the data in the target domain are not labeled, probabilistic predictions of the target data (soft label) are used.

Finally, we design a sample selection strategy based on confident learning [36] to obtain high-confidence labeled samples from the target domain. Confident learning focuses on label quality by characterizing and identifying label errors in a dataset, which uses a probability threshold to estimate noise and rank examples to train with confidence. In our method, the pseudo-labels of the target data are ranked according to their confidence to obtain the high-confidence pseudo-labels. The obtained high-confidence labeled examples of the target domain are used to fine-tune the domain adaptation model to enhance the discrimination of the model for the target domain.

The contributions of our proposed SCLUDA can be summarized as follows.

1) We perform SCL in the source and the target domains to minimize the distance of all positive pairs with respect to negative pairs in the domain to improve the separability of the intradomain data.
2) We introduce a novel DSL based on OCC to project features from the source and target domains into a unified new nominal domain, which makes the distributions of the source and target domains more similar, thereby reducing the discrepancy between domains.
3) A sample selection strategy based on confidence learning is designed to obtain high-confidence labeled samples from the target domain, and then, these samples are used to fine-tune the domain adaptation model, so as to enhance the discrimination of the model for the target domain.

This article is organized into several sections. Section II provides an overview of the recent advancements in UDA and contrastive learning. In Section III, our proposed method is thoroughly explained. Section IV contains a comprehensive analysis and presentation of the results from our experiments. This article is concluded in Section V.

II. Related Work

A. UDA From a Statistical Perspective

UDA is a field of machine learning that aims to transfer knowledge learned from a source domain to a target domain without using labeled data from the target domain [22], [37]. UDA is important because labeled data are often expensive or difficult to obtain.

From a statistical perspective, most related work can be roughly divided into three categories: marginal distribution matching, marginal and class-conditional distribution matching, and joint distribution matching.

Marginal distribution matching refers to the process of aligning the source distribution $P(x^s)$ and the target distribution $P(x^t)$. This type of method is usually based on the assumption that there exists a feature mapping that makes the marginal distributions similar and the labeling distributions similar as well [22], [23], [38]. Pan et al. [22] presented the concept of the MMD for measuring the discrepancy between distributions. They also introduced a technique called transfer component analysis (TCA) that utilizes MMD to learn a set of transfer components in a reproducing kernel Hilbert space (RKHS). The objective of TCA is to minimize the differences in marginal distributions between domains while maximizing the variance of the data. DAN [23] aligns the distributions of two domains by simultaneously minimizing the classification loss on the source domain and the MK-MMD metric between the two domains. While there is theoretical justification for matching the marginal distributions [39], it should be noted that this does not necessarily mean that the labeling distributions are also matched, as mentioned in [40] and [41].

For matching marginal and class-conditional distributions, it is generally assumed that there exists a feature transformation that makes the marginal distributions similar, and the class-conditional distributions similar as well. To achieve this, the transformation can be learned by minimizing both the discrepancy in marginal distributions and the discrepancy in class-conditional distributions using either the Jensen–Shannon divergence [42] or the MMD distance [18], [43]. Long et al. [18] introduced a method called joint distribution adaptation (JDA). JDA aims to align the marginal distribution using the MMD and the class-conditional distribution using a class-conditional MMD by finding a linear transformation. Zhu et al. [43] introduced a novel network called attention-based multiscale residual adaptation network (AMRAN) for cross-scene classification. AMRAN consists of three main modules: a residual adaptation module for aligning marginal distributions, an attention module for extracting robust features, and a multiscale adaptation module for extracting features at multiple scales and aligning class-conditional distributions. While these methods have shown empirical effectiveness, they may not fully achieve the goal of matching the source and target joint distributions. This is because a joint distribution $P(x, y)$ can be decomposed into the product of the marginal distribution and the labeling distribution, i.e., $P(x, y) = P(x)P(y/x)$, rather than the product of the marginal distribution and the class-conditional distribution, i.e., $P(x, y) \neq P(x)P(x/y)$.

Joint distribution matching mainly refers to directly matching between the source joint distribution and the target joint distribution. Courty et al. [41] aimed to reduce the discrepancy in joint distribution by minimizing the optimal transport loss. Chen et al. [44] utilized linear projections to directly align the source and target joint distributions using the L^2-distance.

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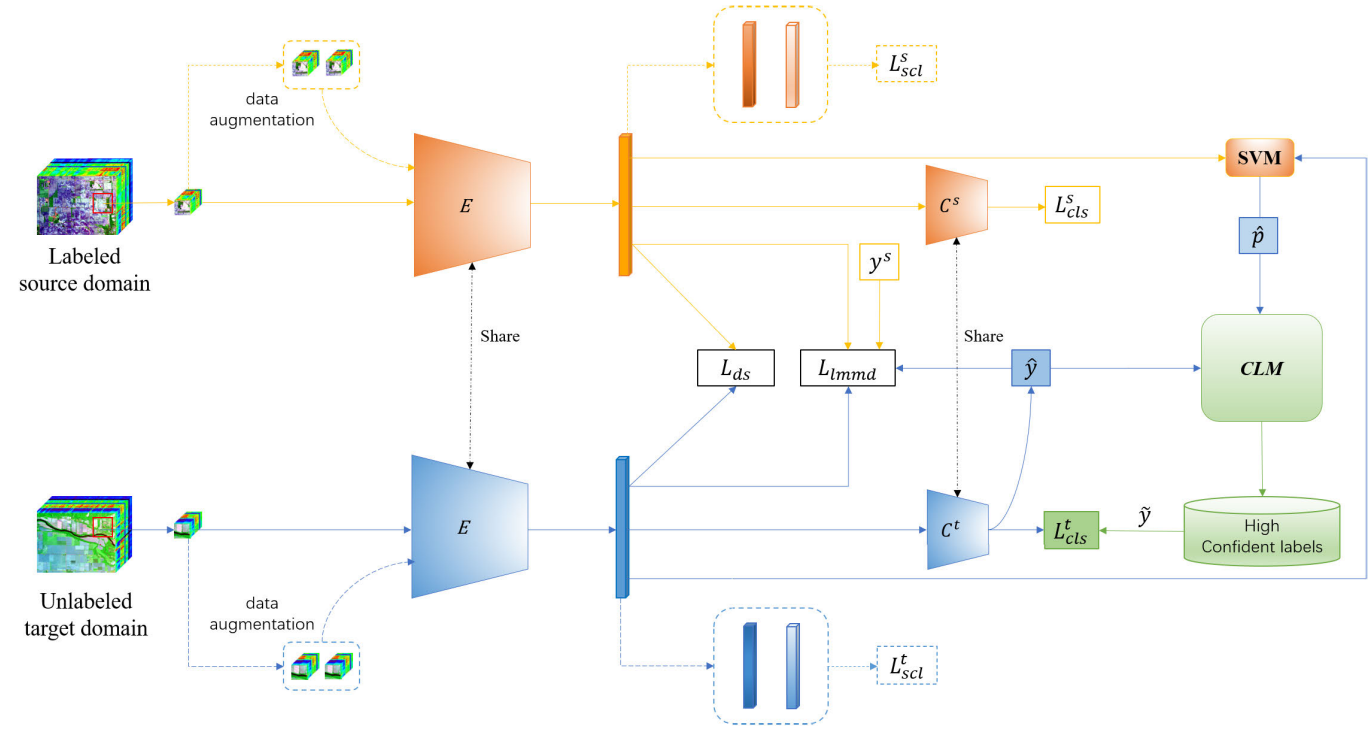


Fig. 1. Overview architecture of the SCLUDA network. The orange solid line and blue solid line, respectively, indicate the computational flow of the source and target samples. The orange dotted line and blue dotted line, respectively, indicate the computational flow of the augmented source and target samples.

B. Contrastive Learning

Contrastive learning has emerged as a powerful approach to learn representations of data by maximizing the similarity between similar data pairs and minimizing the similarity between dissimilar data pairs. This approach has been applied to various tasks in machine learning, including computer vision [32], natural language processing [45], and audio processing [46]. Many researchers [47], [48], [49], [50], [51] have endeavored to comprehend and elucidate its characteristics, along with its impacts on downstream tasks. In recent years, several methods have been proposed to extend contrastive learning to large-scale datasets [32], [52]. They utilize the InfoNCE loss function to bring the positive samples nearer to each other in the embedding space while simultaneously pushing them further away from the negative samples. Supervised contrastive learning [33] is a technique that applies the principles of contrastive learning to the context of fully supervised learning. SCL operates by selecting samples belonging to the same class as positive examples and then aligning them in the embedding space while also pushing away samples from different classes to increase the discriminability of the learned representations. In the HSI field, a refined prototypical contrastive learning network for FSL (RPCL-FSL) [53] is proposed, which integrates supervised contrastive learning and few-shot learning to tackle the challenge of HSI classification. Furthermore, a deep contrastive learning network (DCLN) method [54] is proposed to solve the small-sample classification problem. Much research on contrastive learning has focused on balanced data, and some recent studies [55], [56] have explored the application of contrastive learning to imbalanced and long-tailed classification problems, showing improved performance.

III. Proposed Method

Assume that there is a source domain D_s with labels and a target domain D_t without labels. Specifically, the sample set and the label set are denoted as $X^s = \{x_1^s, x_2^s, \ldots, x_{n_s}^s\}$ and $Y^s = \{y_1^s, y_2^s, \ldots, y_{n_s}^s\}$ in the source domain, respectively, where $y_i^s \in \{1, 2, \ldots, n_c\}$ is the label corresponding to the sample $x_i^s \in R^d$. Similarly, the target domain samples without labels are denoted as $X^t = \{x_1^t, x_2^t, \ldots, x_{n_t}^t\}$. The main objective of UDA is to transfer a classifier model trained by a labeled source domain to target domain. The flowchart of the proposed SCLUDA approach is shown in Fig. 1, which includes four main parts: domain adaptation module, SCL module, confident learning module (CLM), and classification and fine-tuning module. The network structure of the feature extractor E is shown in Fig. 2.

Domain adaptation has to address the challenge of dissimilarities between the data distributions of the source and the target domains. Initially, the feature extractor E is employed to perform forward propagation on the data from the source and target domains, generating domain-invariant features $E(x^s)$ and $E(x^t)$. Next, the extracted features from both domains are input into the classifier to generate the probability of prediction $P^s \in R^{n_c \times n_s}$ and $P^t \in R^{n_c \times n_t}$ for corresponding samples. For the target domain data, the category corresponding to the element with the largest probability is used as the pseudo-label and denoted as $\widehat{Y}^t$. Finally, combined with the extracted features $E(x^s)$ and $E(x^t)$, the corresponding loss functions are calculated to make the distribution of the source and target domains more similar and achieve class-conditional distribution alignment. SCL aims to aggregate samples belonging to the same class in the embedding space while separating samples from different classes. For inputting each sample

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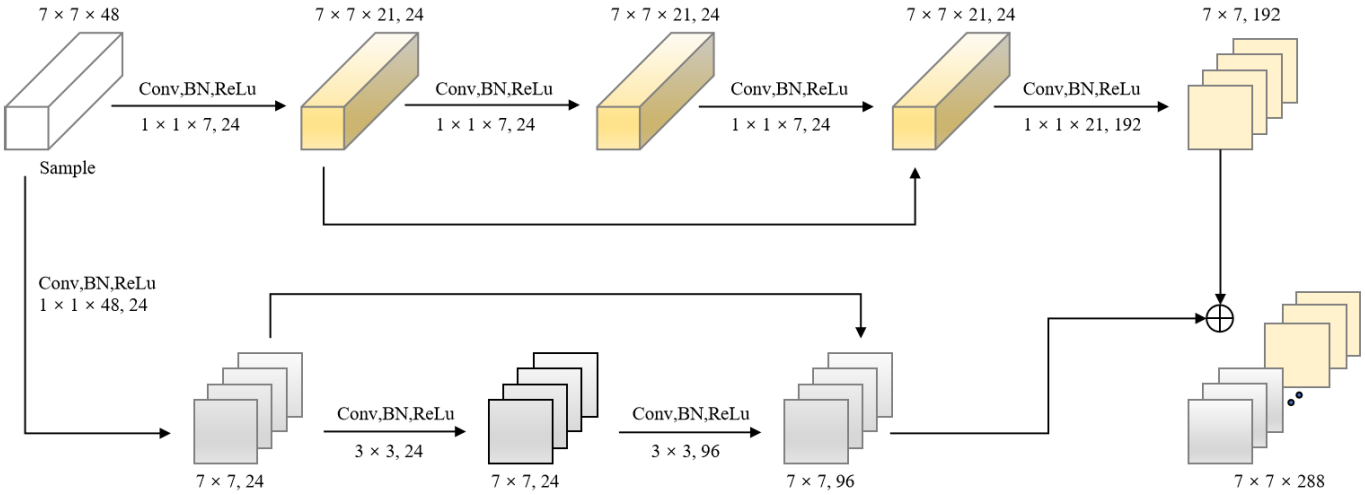


Fig. 2. Network structure of the feature extractor. Take a sample with a size of 7 × 7 × 48 as an example, the top network is responsible for extracting spectral features from the input data, and the bottom network is responsible for extracting spatial features from the input data. ⊕ represents a concatenation operator.

batch, we apply two different data augmentation techniques (such as adding random Gaussian noise and random flip) to obtain two copies of the batch. Both copies are input into the feature extractor E to obtain a 288-D normalized embedding. During training, the resulting features are further propagated to a fully connected layer, whose output is used to calculate the supervised contrastive loss. In the end, to further enhance the discrimination ability of SCLUDA network for target domain data, a CLM is proposed to obtain high-confidence samples and labels. Here, the extracted features $E(x^s)$ from the source domain are used to train an SVM classifier. Then, the features $E(x^t)$ from the target domain are fed to the SVM classifier to obtain predicted probabilities $\widehat{P}$ for the target domain data. The confident joint matrix $C_{\widehat{y},y^t}$ (an unnormalized counts matrix used to partition and count label errors) is calculated by the predicted probabilities $\widehat{P}$ and the pseudo-labels $\widehat{Y}^t$. By normalizing the confident joint matrix $C_{\widehat{y},y^t}$, the joint distribution matrix $Q_{\widehat{y},y^t}$ can be generated, which is used to find all mislabeled examples in the dataset. After that, we perform a ranking and pruning operation to separate the pseudo-labels into two categories: confident labels $\tilde{Y}$ and unreliable labels. In turn, the target samples with high confident labels $\tilde{Y}$ are used to fine-tune the feature extractor E and classifier C to make the classifier more accurate. The total loss function of an SCLUDA is defined as

$$L_{\text{total}} = L_{\text{cls}} + L_{\text{da}} + L^s_{\text{scl}} + L^t_{\text{scl}} \tag{1}$$

where the first term represents the classification loss on the labeled source data and the classification loss on the selected high-confidence labeled target data, the second term represents the loss of domain adaptation, and the third and fourth terms are about the SCL loss of source and target domain data, respectively.

A. Domain Adaptation

In order to address the challenges of domain shift and minimize the discrepancy between domains, the SCLUDA method proposed in this article emphasizes both the similarity of the distribution between the source and target domains and the alignment of class-conditional distributions between domains. During the domain adaptation process, the feature extractor E is trained using both the DSL [34] and the LMMD loss [35]. On one hand, the DSL enables global domain adaptation by making the overall distribution of the two domains to be as similar as possible. On the other hand, the LMMD loss facilitates alignment of the subdomain distribution of the two domains, allowing the deep network to learn more transferable representations. The loss of domain alignment is defined as

$$L_{\text{da}} = \alpha L_{\text{ds}} + \beta L_{\text{lmmd}} \tag{2}$$

where the first term represents DSL, the second term represents the LMMD loss, and α and β are weighted parameters of two losses.

1) Domain Similarity Loss: To achieve domain-level alignment and to minimize the discrepancy between domains, we project the source and target domain data onto a unified new nominal domain, which can be formulated as an OCC problem [57]. Specifically, we process the features $E(x^s)$ and $E(x^t)$ from the source and target domains using a 288 × 1 fully connected layer and a sigmoid activation function, which is able to predict the probability $p^o(x)$ of each sample x belonging to the new nominal domain. The DSL based on OCC is defined as

$$\begin{aligned} L_{ds} &= -y^o \cdot \sum_{i=1}^{n_s} \log p^o\left(x_i^s\right) - y^o \cdot \sum_{i=1}^{n_t} \log p^o\left(x_i^t\right) \\ &= -y^o \cdot \left(\sum_{i=1}^{n_s} \log p^o\left(x_i^s\right) + \sum_{i=1}^{n_t} \log p^o\left(x_i^t\right) \right) \\ &= -y^o \cdot \sum_{j=1}^{n_s+n_t} \log p^o\left(x_j\right) \end{aligned} \tag{3}$$

where y^o represents the domain label and its value is set to 1. The DSL L_{ds} is essentially a cross-entropy loss in an OCC

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task, and by minimizing the DSL, the features of the source domain and target domain can be projected into a unified new nominal domain. By optimizing the domain similar loss, the distribution of the two domains can be more similar to reduce domain discrepancy. Unlike adversarial domain adaptation methods, which confuse the domain discriminator by adding a GRL to achieve similar distributions between domains, the DSL aims to map the features of the source and target domains to a single nominal domain to achieve similarity in distribution between domains and thus has stronger constraints.

2) LMMD Loss: If we only focus on global adaptation and ignore the correlation between subdomains belonging to the same category across different domains, it may result in some irrelevant data being too close together, which can negatively affect the accuracy of classification. To overcome the limitations of global domain adaptation, it is necessary to also employ subdomain adaptation techniques. According to the category of samples, we divide source domain D_s and target domain D_t into n_c subdomains $D_s^{(c)}$ and $D_t^{(c)}$ where $c \in \{1, 2, \ldots, n_c\}$ indicates the class label. Let $p^{(c)}$ and $q^{(c)}$ denote the distribution of $D_s^{(c)}$ and $D_t^{(c)}$, respectively. Based on the idea of MMD method, the LMMD loss can be expressed as

$$d_{\mathcal{H}}(p, q) \triangleq \mathbf{E}_c \left\| \mathbf{E}_{p^{(c)}}\left[\phi\left(x^s\right)\right] - \mathbf{E}_{q^{(c)}}\left[\phi\left(x^t\right)\right] \right\|_{\mathcal{H}}^2 \tag{4}$$

where $\mathcal{H}$ represents an RKHS defined by characteristic kernel k. Here, the mapping function $\phi(\cdot)$ defines a mapping from the original data to RKHS, and the kernel function is defined as the inner product of the map as follows:

$$k\left(x^s, x^t\right) = \left\langle \phi\left(x^s\right), \phi\left(x^t\right) \right\rangle \tag{5}$$

where $\langle \cdot, \cdot \rangle$ represents the inner product operation and x^s and x^t are instances from $D_s^{(c)}$ and $D_t^{(c)}$. Assuming that the weight of each sample belonging to each category is w^c, we present a formulation for an unbiased estimator of (4) as follows:

$$\hat{d}_{\mathcal{H}}(p, q) = \frac{1}{n_c} \sum_{c=1}^{n_c} \left\| \sum_{x_i^s \in \mathcal{D}_s} w_i^{sc} \phi\left(x_i^s\right) - \sum_{x_j^t \in \mathcal{D}_t} w_j^{tc} \phi\left(x_j^t\right) \right\|_{\mathcal{H}}^2 \tag{6}$$

where w_i^{sc} and w_j^{tc} represent the weight of the instances x_i^s and x_j^t belonging to category c, respectively. It is worth noting that both $\sum_{i=1}^{n_s} w_i^{sc}$ and $\sum_{j=1}^{n_t} w_j^{tc}$ are equal to 1, and $\sum_{x_i \in \mathcal{D}} w_i^c \phi(x_i)$ represents the result of a weighted summation on class c. The weight w^c is defined as

$$w_i^c = \frac{y_i^c}{\sum_{(x_j, y_j) \in \mathcal{D}} y_{jc}} \tag{7}$$

where y_i^c is the cth of vector y_i. Since the source domain data has a real label, we convert the true label y_i^s into a one-hot vector to further calculate the weight w_i^{sc} for each sample. However, in UDA, the target domain data have no available label information. As we all know, the output of the classifier is a possibility distribution, which shows that the possibility of the sample x_i^t belongs to each category. Thus, we use the output of the classifier to calculate the weight w_j^{tc} for each target sample. In addition, since we cannot directly calculate $\phi(\cdot)$, we rewrite (6) as

$$\begin{aligned} L_{\text{lmmd}} = \frac{1}{n_c} \sum_{c=1}^{n_c} \Bigg[& \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} w_i^{sc} w_j^{sc} k\left(\mathbf{z}_i^{\text{sl}}, \mathbf{z}_j^{\text{sl}}\right) \\ & + \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} w_i^{tc} w_j^{tc} k\left(\mathbf{z}_i^{\text{tl}}, \mathbf{z}_j^{\text{tl}}\right) \\ & - 2 \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} w_i^{sc} w_j^{tc} k\left(\mathbf{z}_i^{\text{sl}}, \mathbf{z}_j^{\text{tl}}\right) \Bigg]. \end{aligned} \tag{8}$$

It is worth noting that when the weights of LMMD are equal, it becomes a general form of conditional MMD (CMMD).

B. Supervised Contrastive Learning

SCL [33] aims to maximize the consistency between differently augmented views of samples from the same category, thereby reducing the distance between samples from the same category while increasing the distance between samples from different categories. As a result, the SCLUDA network is able to better distinguish between samples from the source domain and those from the target domain. The general framework of contrastive learning comprises four main components: a data augmentation module, an encoder network, a projection head network, and a supervised contrastive loss function.

1) Data Augmentation Module, $Aug(\cdot)$: For each sample x in the source and target domain, data augmentation is performed twice to obtain two related views of the same sample, $\tilde{x} = \text{Aug}(x)$. Within this article, we have opted to employ two frequently utilized techniques for data augmentation: random Gaussian noise and random flip.

2) Encoder Network, $E(\cdot)$: The augmented data are fed into the feature extractor E in order to derive a representation vector, $\boldsymbol{r} = E(x) \in \mathcal{R}^{D_E}$, where $D_E = 288$ in our experiments. The resulting vector $\boldsymbol{r}$ is then normalized to the unit hypersphere in $\mathcal{R}^{D_E}$.

3) Projection Head Network, $Pro(\cdot)$: The vector $\boldsymbol{r}$ is then passed through the projection head network to obtain the vector $\boldsymbol{z} = \text{Pro}(\boldsymbol{r}) \in \mathcal{R}^{D_P}$, where $D_P = 128$ in our experiments. For this task, we have opted to utilize a singular linear layer as the projection head network. Similarly, we apply a normalization process to the output of this network, constraining it to lie on the unit hypersphere. This normalization facilitates the use of inner products as a measure of distances within the projection space.

4) Supervised Contrastive Loss: Take the source domain data as an example. We randomly select N sample/label pairs, $\{x_k^s, y_k^s\}_{k=1,\ldots,N}$. Through the data augmentation module, $2N$ sample/label pairs are obtained, denoted as $\{\tilde{x}_l^s, \tilde{y}_l^s\}_{l=1,\ldots,2N}$, where $\tilde{x}_{2k}^s$ and $\tilde{x}_{2k-1}^s$ are two views of x_k^s, $\tilde{y}_{2k-1}^s = \tilde{y}_{2k}^s = y_k^s$. Supervised contrastive loss for source domain is defined as

$$L_{\text{scl}}^s = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\left(\boldsymbol{z}_i^s \cdot \boldsymbol{z}_p^s / \tau\right)}{\sum_{a \in A(i)} \exp\left(\boldsymbol{z}_i^s \cdot \boldsymbol{z}_a^s / \tau\right)} \tag{9}$$

where $i \in I \equiv \{1, \ldots, 2N\}$ is the index of an augmented sample called the anchor, $\boldsymbol{z}_l^s = \text{Pro}(E(\tilde{x}_l^s)) \in R^{D_P}$, the $\cdot$

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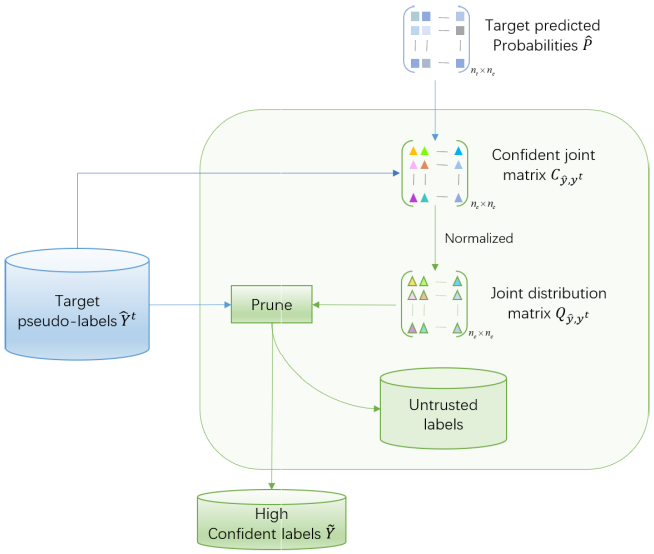


Fig. 3. Flowchart of CLM. $\hat{Y}^t$ represents the pseudo-label of the target domain, and $\hat{P}$ represents the target predicted probabilities from SVM. First, calculate the confidence joint matrix $C_{\hat{y},y^t}$ through formula (11). Second, calculate the joint distribution matrix $Q_{\hat{y},y^t}$ through formula (13). Finally, select high-confidence samples through the rank and prune method.

symbol represents the inner product, $A(i) \equiv I \setminus \{i\}$, $P(i) \equiv \{p \in A(i) : \tilde{\boldsymbol{y}}_p^s = \tilde{\boldsymbol{y}}_i^s\}$ is the collection of all positives, $|P(i)|$ is its cardinality, and $\tau \in R^+$ is a temperature parameter. The gradient exhibits an inverse relationship with the choice of parameter τ. Therefore, we adjust the loss through τ to ensure stability during training. Similarly, the supervised contrastive loss for target domain is defined as

$$L_{scl}^t = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp\left(z_i^t \cdot z_p^t / \tau\right)}{\sum_{a \in A(i)} \exp\left(z_i^t \cdot z_a^t / \tau\right)}. \quad (10)$$

The output of a deep neural network is a probability distribution that nicely characterizes the probability of assigning x_i to each class C. Hence, when dealing with an unlabeled target domain D_t, it is intuitive to utilize the output y_i^t of the network as the probability for assigning x_i^t to each class C.

C. Confident Learning Module

The objective of confident learning [36] is to assess the joint distribution of noisy and true labels, with the goal of identifying all incorrectly labeled samples. In this article, the CLM comprises two inputs: the predicted probabilities $\hat{P}$ from the SVM classifier trained on source samples and the noisy pseudo-labels $\hat{Y}^t$ from the classifier C. Our strategy based on confident learning involves two main steps: 1) calculate the joint distribution matrix between the pseudo-labels (represented using $\hat{Y}^t$) and the true labels (which are unknown and approximated using $\hat{P}$) to assess the confidence of the pseudo-labels and 2) filter out untrusted samples and extract reliable samples. The flowchart of CLM is shown in Fig. 3.

First, the confident joint $C_{\hat{y},y^t}$ is a matrix recording the number of correctly labeled and incorrectly labeled samples. As an example, $C_{\hat{y}=5,y^t=2} = 10$ means that the pseudo-label of the ten samples is 5, but the true label is 2. The diagonal entries of the matrix capture the number of correctly labeled instances, while the nondiagonal entries represent the number of instances with labeling errors. However, effectively obtaining the true labels of the target domain is not possible, so the confident joint matrix $C_{\hat{y},y^t}$ is obtained by evaluating the predicted probabilities $\hat{P}$ and pseudo-labels $\hat{Y}^t$. Specifically, $C_{\hat{y},y^t}$ estimates $\boldsymbol{X}^t_{(\hat{y}=i,y^t=j)}$, the set of target samples x^t that have the pseudo-label i and actually have true label j, by dividing x^t into estimate bins $\hat{\boldsymbol{X}}^t_{(\hat{y}=i,y^t=j)}$. Here, $\hat{\boldsymbol{X}}^t_{(\hat{y}=i,y^t=j)}$ is another set of target samples x^t with noisy label $\hat{y} = i$ that have large enough predicted probabilities $\hat{p}(\hat{y} = j; x^t)$ for class $y^t = j$, determined by a per-class threshold t_j. It is proved that when $\hat{\boldsymbol{X}}^t_{(\hat{y}=i,y^t=j)} = \boldsymbol{X}^t_{(\hat{y}=i,y^t=j)}$, $C_{\hat{y},y^t}$ can effectively identify labeling errors. The matrix $C_{\hat{y},y^t}$ is defined as

$$\begin{aligned} C_{\hat{y},y^t}[i][j] &:= \left|\widehat{X}^t_{(\hat{y}=i,y^t=j)}\right| \quad \text{where} \\ \widehat{X}^t_{(\hat{y}=i,y^t=j)} &:= \{x^t \in \boldsymbol{X}^t_{(\hat{y}=i)} : \hat{p}\left(\hat{y}=j; x^t\right) \geq t_j, \\ j &= \underset{l \in \{1,\ldots n_c\}: \hat{p}\left(\hat{y}=l; x^t\right) \geq t_l}{\operatorname{argmax}} \hat{p}\left(\hat{y}=l; x^t\right)\} \end{aligned} \quad (11)$$

$$t_j = \frac{1}{\left|X^t_{(\hat{y}=j)}\right|} \sum_{x^t \in X^t_{(\hat{y}=j)}} \hat{p}\left(\hat{y}=j; x^t\right) \quad (12)$$

where t_j is the average self-confidence for each category. Here, a definition of self-confidence is introduced, which represents the predicted probability that the sample x^t belongs to its given label $\hat{y}$, represented as $\hat{p}(\hat{y} = j; x^t \in \boldsymbol{X}_{t(\hat{y}=j)})$. Then, by normalizing $C_{\hat{y},y^t}$, $Q_{\hat{y},y^t}$ is estimated as

$$\widehat{\boldsymbol{Q}}_{\hat{y}=i,y^t=j} = \frac{\frac{C_{\hat{y}=i,y^t=j}}{\sum_{j \in \{1,\ldots,n_c\}} C_{\hat{y}=i,y^t=j}} \cdot \left|X^t_{(\hat{y}=i)}\right|}{\sum_{i,j \in \{1,\ldots,n_c\}} \left(\frac{C_{\hat{y}=i,y^t=j}}{\sum_{j \in \{1,\ldots,n_c\}} C_{\hat{y}=i,y_t=j}} \cdot |X^t_{(\hat{y}=i)}|\right)}. \quad (13)$$

Second, according to the estimation of $C_{\hat{y},y^t}$ and $Q_{\hat{y},y^t}$, the prune by class (PBC) method [36] is used to clean data. For each class $i \in \{1, \ldots, n_c\}$, the target samples are sorted according to their self-confidence $\hat{p}(\hat{y} = j; x^t \in \boldsymbol{X}^t_{(\hat{y}=j)})$. A confidence sample set $\boldsymbol{X}_{\text{confidence}}$ is obtained by pruning the $n_t \cdot \sum_{j \in \{1,\ldots,K\}: j \neq i} (\widehat{\boldsymbol{Q}}_{\hat{y}=i,y^t=j}[i])$ samples with the lowest self-confidence. In order to obtain a higher confidence sample set $\boldsymbol{X}_{\text{high-confidence}}$, a selection strategy is represented as

$$\boldsymbol{X}_{\text{high-confidence}} = \left\{x^t \in \boldsymbol{X}_{\text{confidence}} \mid \max\left\{\hat{p}\left(x^t\right)\right\} \geq T\right\} \quad (14)$$

where T is a threshold parameter.

D. Classification and Fine-Tuning Module

In the SCLUDA network, we train the classifier C by utilizing labeled data from the source domain as well as labeled data with high confidence in the target domain. As both the source feature $E(x^s)$ and the target feature $E(x^t)$ are domain-invariant features, the classifier trained using the source data can be utilized directly to classify the target data within the shared feature space. The feature extractor E is trained to be more discriminative by incorporating the classification loss on the labeled source data during training. Moreover, we further optimize the SCLUDA network by fine-tuning it with high-confidence labeled data from the target domain, enhancing its ability to discriminate against samples within the

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target domain. However, since a high-confidence label is not equivalent to a ground-truth label, we introduce a weight γ to restrict the classification loss of the target domain. In summary, the classification loss can be represented as

$$L_{\text{cls}} = L^s_{\text{cls}} + \gamma L^t_{\text{cls}}. \tag{15}$$

1) Source Classification Loss: Cross-entropy loss is a commonly used loss function in deep neural networks for multicategory classification tasks in supervised learning. The classification loss of the source domain can be represented as

$$L^s_{\text{cls}} = \frac{1}{n_s}\sum_{i=1}^{n_s} L_{\text{cross-entory}}\left(C\left(E\left(x^s_i\right)\right), y^s_i\right) \tag{16}$$

where $C(\cdot)$ represents the predicted value of the classifier C.

2) Target Fine-Tuning Loss: Similar to the source domain classification loss, the feature extractor E and classifier C are fine-tuned through a cross-entropy loss using the obtained high-confidence sample labels $\tilde{Y}$. The loss function is defined as

$$L^t_{\text{cls}} = \frac{1}{n_t}\sum_{i=1}^{n_t} L_{\text{cross-entory}}\left(C\left(E\left(x^t_i\right)\right), \tilde{y}_i\right). \tag{17}$$

E. Implementation Details

In order to select high-confidence target sample labels and improve the classification accuracy on target domain, the training process of the SCLUDA is divided into two stages. The total training procedure is detailed in Algorithm 1. At the start of the training process, since the SCLUDA network lacks effective discrimination ability for target samples, we first train feature extractors E and classifiers C through domain adaptation and SCL. The total loss at this stage is expressed as

$$\min_{\theta_e,\theta_c} L^s_{\text{cls}} + L_{\text{da}} + L^s_{\text{scl}} + L^t_{\text{scl}} \tag{18}$$

where θ_e and θ_c are the parameters of E and C, respectively.

After TRAIN_NUM epochs, the classifier C has acquired a certain degree of discriminative ability toward the target samples. By leveraging CLM, we identify high-confidence target samples in each ITER_NUM epoch and utilize them for fine-tuning the deep network, thereby further boosting the performance of the classifier. The loss of fine-tuning can also be expressed as

$$\min_{\theta_e,\theta_c} \gamma L^t_{\text{cls}}. \tag{19}$$

Algorithm 1 Algorithm of SCLUDA for UDA

Input: Source domain samples $\boldsymbol{X}^s$ and labels $\boldsymbol{Y}^s$, target domain samples $\boldsymbol{X}^t$, hyperparameter α, β and γ, the total epoch number NUM, the epoch number to start confident learning $TRAIN_NUM$, the interval epoch number to reset confidenct learning $ITER_NUM$.

Output: The parameters θ_e, θ_c of feature extractor E and classifier C. Randomly initialize θ_e, θ_c and set the values of α, β, γ.

```
for epoch in {1, ..., NUM} do
    X^s1, X^t1 = Augment1(X^s, X^t)
    X_s2, X_t2 = Augment2(X^s, X^t)
    (X^s, Y^s) → L^s_cls
    (X^s, X^t) → L_ds
    Regard the network output as the target domain sample pseudo-labels Ŷ^t
    (X^s, Y^s, X^t, Ŷ^t) → L_lmmd
    (X^s1, X^s2, Y^s) → L^s_scl
    (X^t1, X^t2, Ŷ^t) → L^s_scl
    if epoch ≥ TRAIN_NUM then
        if epoch%ITER_NUM = 0 then
            Obtain target predicted probabilities P̂
            and the target pseudo labels Ŷ^t
            Calculate the confident joint matrix C_{ŷ,y^t}
            through Eq.(11)
            Calculate the joint distribution matrix Q_{ŷ,y^t}
            through Eq.(13)
            Obtain high-confidence target sample labels Ỹ
        end if
        Update θ_e, θ_c through Eq.(18)
        Fine-tune θ_e, θ_c through Eq.(19)
    else
        Update θ_e, θ_c through Eq.(18)
    end if
end for
```

IV. Experimental Results

A. Datasets

To attest to the potency of our methods, we conducted experiments on six HSI datasets, including Pavia University and Pavia Center, Houston2013 and Houston2018, as well as Shanghai and Hangzhou.

1) Pavia University and Pavia Center: Both datasets were procured through the reflective optics spectrographic Image system (ROSIS) sensors in Pavia, which is located in the northern region of Italy. The Pavia University contains 610 × 610 pixels. Initially, the dataset consisted of 115 spectral bands, but after the removal of 12 noisy bands, the remaining dataset comprised 103 spectral bands. The Pavia Center dataset includes 1096 × 1096 pixels and 102 spectral bands. Due to the presence of some invalid data within the aforementioned datasets, the size of the former was reduced to 610 × 315 pixels, while the size of the latter was adjusted to 1906 × 715 pixels after removing the invalid data. In order to ensure uniformity, the last spectral band of the Pavia University dataset was removed. Fig. 4 shows the false-color images and ground-truth maps for both datasets.

2) Houston2013 and Houston2018: The two datasets, which encompass the University of Houston campus and the adjoining regions, were obtained by distinct sensors in 2013 and 2018, respectively. The Houston2013 dataset has a spatial size of 349 × 1905 pixels and a spatial resolution of 2.5 m. It contains 144 spectral bands for analysis and includes 15 distinct categories. The Houston2018 dataset has a size of 209 × 955 pixels and a spatial resolution of 1 m. It comprises 48 spectral bands and encompasses a total of 20 categories.

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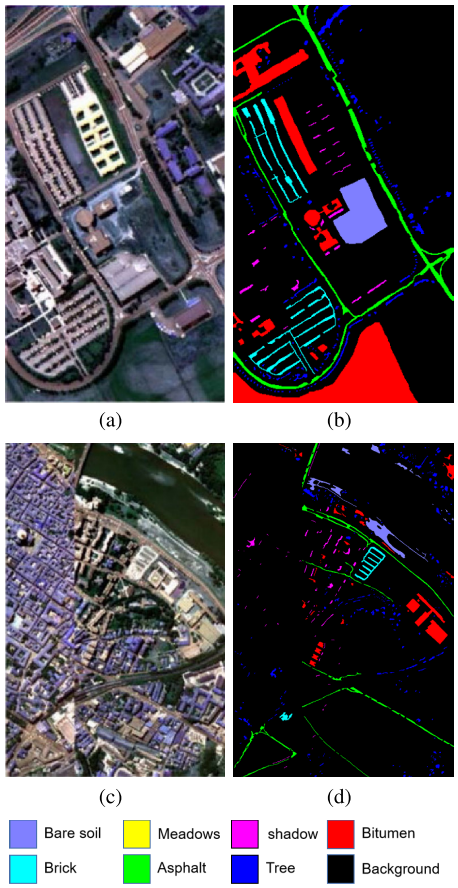


Fig. 4. Pavia image. (a) False-color image of UP. (b) Ground-truth map of UP. (c) False-color image of PC. (d) Ground-truth map of PC.

The wavelength range of both datasets spans from 380 to 1050 nm. To ensure consistency, 48 spectral bands and 209 × 955 overlapping areas were selected from the Houston2013 dataset. Fig. 5 shows the false-color images and ground-truth maps for both datasets.

3) Shanghai and Hangzhou: The Shanghai and Hangzhou datasets were collected using the EO-1 Hyperion hyperspectral sensor, resulting in 198 bands after removal of bad bands. The Shanghai dataset comprises 1600 × 230 pixels, while the Hangzhou dataset has 590 × 230 pixels. Both datasets have three categories in common, namely, water, ground/buildings, and plants. Fig. 6 shows the false-color images and ground-truth maps for both datasets.

Using the six hyperspectral datasets, we formulated three domain adaptation tasks, which are delineated as follows.

4) Task 1 (Pavia Task): The Pavia University dataset is considered as the source domain and the Pavia Center dataset as the target domain. The task considers seven categories, which are shown in Table I.

5) Task 2 (Houston Task): The Houston2013 dataset is considered as the source domain and the Houston2018 dataset as the target domain. In order to accomplish this task, seven common categories are considered in both datasets. Table II presents the detailed information pertaining to the datasets used in our study.

6) Task 3 (Shanghai–Hangzhou Task): The Shanghai dataset is used as the source domain and the Hangzhou dataset as the target domain. The three common categories present in both datasets are considered in this task, which is presented in Table III.

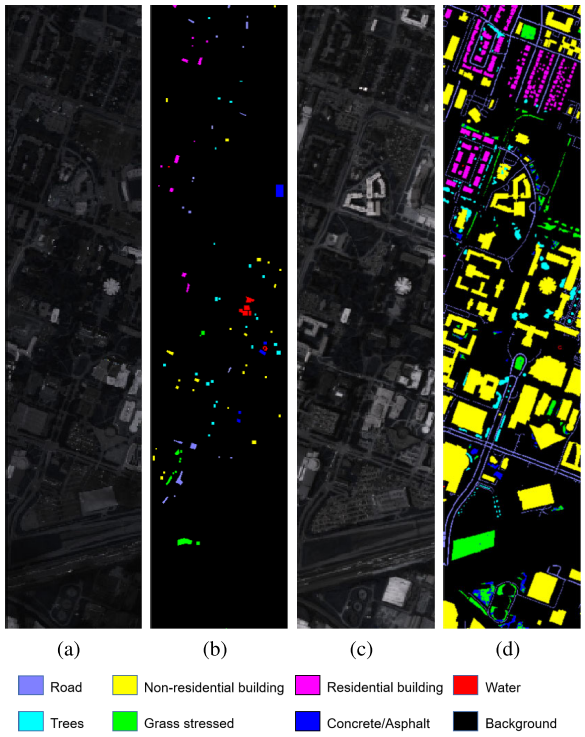


Fig. 5. Houston image. (a) False-color image of Houston 2013. (b) Ground-truth map of Houston 2013. (c) False-color image of Houston 2018. (d) Ground-truth map of Houston 2018.

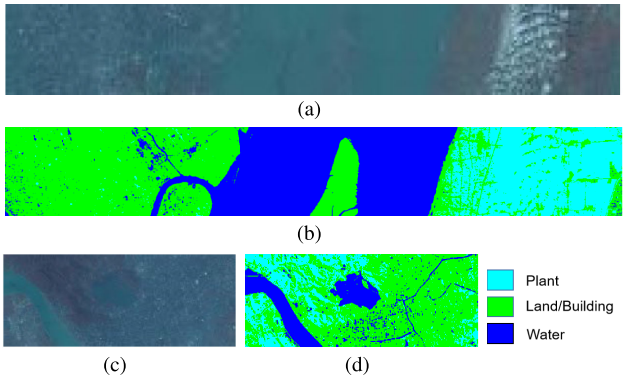


Fig. 6. Shanghai–Hangzhou image. (a) False-color image of Shanghai. (b) Ground-truth map of Shanghai. (c) False-color image of Hangzhou. (d) Ground-truth map of Hangzhou.

TABLE I
NUMBER OF SAMPLES IN THE PAVIA TASK

PaviaU and PaviaC data			
No.	Class	PaviaU (Source)	PaviaC (Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

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TABLE II
NUMBER OF SAMPLES IN THE HOUSTON TASK

Houston2013 and Houston2018 data			
No.	Class	Houston2013 (Source)	Houston2018 (Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE III
NUMBER OF SAMPLES IN THE SHANGHAI–HANGZHOU TASK

Shanghai-Hangzhou data			
No.	Class	Shanghai (Source)	Hangzhou (Target)
1	Water	123123	18043
2	Land/building	161689	77450
3	Plant	83188	40207
Total		368000	135700

B. Experimental Setup

In order to assess the efficacy of our proposed method, we compared it with seven other domain adaptation methods, which include TSVM [16], DAN [23], DANN [27], ED-DMM-UDA [29], CDA [28], CLDA [30], and TAADA [31]. For the experiments, we selected 180 samples for each class in the source domain and used all of the target samples for training. Also, all experiments are conducted ten times to eliminate the influence of random sampling, and the average value is used as the final classification result. To minimize the distribution shift between the source and target domains, we normalized the source and target domain data prior to training.

For TSVM, the Cl is set as 1.5 and the Cu is set as 0.001. For DAN, ResNet-50 is used as the feature extractor. In particular, three adaptive layers are added simultaneously, and each layer uses the MK-MMD metric. The number of different Gaussian kernels is set to 5. The iteration is set to 1000, the batch size is set to 32, and the learning rate is set to 0.01. The optimizer used is stochastic gradient descent (SGD) with a momentum of 0.9 and a weight decay of 0.0005. The input sizes in the three tasks are set to 9×9, 5×5, and 1×1. For DANN, a ResNet18 architecture is utilized as the feature extractor, while the domain classifier is composed of three fully connected layers designed to discriminate between the source and target domains. The batch size, the learning rate, and the epoch are the same as DAN. The input sizes in the three tasks are set to 7×7, 7×7, and 1×1. For ED-DMM-UDA, the source epoch is set as 1000, the target epoch is set as 20, and the input sizes are set as 5×5 in Pavia and Houston tasks. In the Shanghai–Hangzhou task, the input size is set as 1×1. For CDA, the batch size is set to 128 and the learning rates are set to 0.001, 0.05, and 0.005 in three tasks. The ratios of $\lambda 1$ and $\lambda 2$ are set as 10:1, 1:1, and 1:1. The weights are set as 0.5, 0.5, and 0.7. For TAADA, the epoch is set as 100. The other parameters of all methods are uniform and consistent with the original paper.

For our proposed method, we set the total number of epochs as 100. During the training process, we used the mini-batch SGD optimizer for backpropagation, with a weight decay of 0.0005 and a momentum of 0.9. The learning rate η_0 is set as 0.01, 0.001, and 0.0003 for three tasks. In Task 1, the learning rate is adjusted dynamically during training using the following formula: $\eta_\theta = \eta_0/(1 + a\theta)^b$, where $a = 10$, $b = 0.75$, and θ is a variable that changes linearly from 0 to 1 during the training process. The patch size is set as 7×7, 11×11, and 1×1 in three tasks. Moreover, the batch size of 32 has been designated, and TRAIN_NUM is established as 40. Commencing from epochs = TRAIN_NUM, execute a confidence operation every 20 epochs. Experiments in SCL [33] show that when the parameter τ is set to 0.1, the effect is the best. Therefore, we also set the parameter τ to 0.1. The threshold T for CLM is set to 0.8.

To assess the effect of various methods on domain adaptation tasks, the overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) are employed for verification purposes.

C. Experiments

1) Experiments on Pavia Task: Table IV presents the experimental outcomes of diverse comparative algorithms on the Pavia dataset. The findings indicate that all deep learning-based algorithms outperform the traditional TSVM algorithm in terms of performance. This observation further suggests that deep neural networks are capable of extracting more discriminative features compared to traditional algorithms, leading to improved performance.

Among the deep learning algorithms, DAN solely employing MMD and DANN solely utilizing domain discriminant techniques exhibit inferior performance compared to CDA, CLDA, and TAADA algorithms. The former methods primarily focus on the marginal distribution of the two domains while disregarding the data category information and spectral–spatial characteristics of hyperspectral imaging data. The CDA method achieves classwise distribution alignment by integrating classwise adversarial adaptation with classwise probability MMD (PMMD) techniques. As a result, CDA exhibits a significant improvement of 3.85% in OA compared to DAN and a noteworthy 2.35% increase compared to DANN. Both CLDA and TAADA adopt the idea of adversarial domain adaptation based on two classifiers. In CLDA, the neural network is enhanced through confident learning, where samples with high-confidence false labels are utilized to update the network. This approach further boosts the discriminative capability of the network. The experimental results reveal that CLDA surpasses DAN by 14.82% and DANN by 13.32% in terms of OA. TAADA employs a two-branch attention network to extract spectral–spatial information from HSI, which facilitates the extraction of domain-invariant features. The experimental results demonstrate that TAADA outperforms DAN by 12.29% and DANN by 10.79% in terms of OA. Our proposed method simultaneously considers the similarity of overall distribution between domains, class-conditional distribution alignment, and spectral–spatial information of HSI. As a result, our method yields superior performance compared to other domain

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TABLE IV
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA TASK

Class	TSVM [16]	DAN [23]	DANN [27]	ED-DMM-UDA [29]	CDA [28]	CLDA [30]	TAADA [31]	SCLUDA
1	**98.55**	72.75	75.09	98.21	92.02	96.58	88.78	94.81
2	96.73	82.85	78.23	96.63	77.63	**99.68**	98.62	99.02
3	69.96	77.15	80.69	72.40	86.90	78.07	65.74	**94.36**
4	32.19	67.06	73.08	67.94	78.17	80.07	72.05	**85.03**
5	**100**	98.25	98.28	**100**	99.94	99.99	99.90	97.88
6	23.02	73.09	86.69	25.46	61.81	86.96	94.65	**96.60**
7	86.39	75.17	73.51	83.92	71.60	90.59	**96.30**	90.70
OA	76.03 ±0.02	76.66 ±1.41	78.16 ±4.08	82.50 ±0.50	80.51 ±1.02	91.48 ±2.26	88.95 ±2.82	**93.63 ±0.34**
AA	72.41 ±2.22	78.05 ±1.17	80.79 ±2.52	77.80 ±0.56	81.15 ±1.81	90.28 ±1.96	88.01 ±3.31	**94.06 ±0.88**
$\kappa \times 100$	70.70 ±3.05	72.25 ±1.61	74.16 ±4.62	78.73 ±0.54	76.68 ±1.22	89.71 ±2.72	86.76 ±3.33	**92.36 ±0.40**

TABLE V
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE HOUSTON TASK

Class	TSVM [16]	DAN [23]	DANN [27]	ED-DMM-UDA [29]	CDA [28]	CLDA [30]	TAADA [31]	SCLUDA
1	**99.70**	81.79	74.13	68.14	95.08	63.03	88.99	88.94
2	27.04	76.41	74.69	**85.45**	51.96	84.85	48.69	69.43
3	24.33	70.87	68.62	57.86	62.52	**72.18**	60.38	64.97
4	92.73	**98.18**	96.59	79.55	95.45	94.55	93.18	97.27
5	81.36	81.32	76.64	**97.77**	70.35	93.59	77.84	86.62
6	49.27	47.76	57.76	56.62	**82.74**	62.16	76.68	77.85
7	62.39	56.25	49.51	75.97	26.29	**79.26**	77.95	69.31
OA	52.03 ±0.32	56.87 ±1.48	61.23 ±4.65	66.09 ±1.83	71.18 ±1.31	70.02 ±2.94	73.85 ±3.57	**76.56 ±1.83**
AA	62.40 ±1.30	73.23 ±1.40	71.14 ±3.50	74.48 ±1.66	69.20 ±3.16	78.53 ±2.06	74.82 ±2.64	**79.20 ±1.24**
$\kappa \times 100$	34.29 ±0.80	42.51 ±1.56	45.21 ±4.60	53.10 ±2.18	51.85 ±3.23	57.44 ±3.11	60.39 ±4.25	**63.98 ±2.19**

adaptation methods, achieving an outstanding OA of 93.63%. Fig. 7 shows the classification maps obtained from all the algorithms.

2) Experiments on Houston Task: The Houston task is particularly challenging due to several factors, such as datasets acquired from different sensors and years, and the presence of similar categories (e.g., healthy and stressed grass). These factors contribute to the complexity and difficulty of the domain adaptation task. Despite the challenges posed by the discrepancy in datasets, our methods have demonstrated outstanding performance in comparison to other domain adaptation methods. Table V presents the experimental results of various comparison methods on the Houston data for performance evaluation. The methods, such as ED-DMM-UDA, CDA, CLDA, and TAADA, which consider multiple factors, still demonstrate superior performance compared to methods such as DAN and DANN, which only consider a single factor. In particular, ED-DMM-UDA leverages an adversarial approach to extract domain-invariant features from both the source and target domain samples. Meanwhile, it also encourages the target feature embeddings and the source feature embeddings to form a similar cluster. The experimental results demonstrate that ED-DMM-UDA outperforms DAN and DANN by 9.22% and 4.86%, respectively, in terms of OA. Notably, in contrast to ED-DMM-UDA, our approach uses SCL to bring the clusters of samples belonging to the same class together in the embedding space while simultaneously pushing the clusters of samples belonging to different classes apart. Our method of classification accuracy reached 76.56%. The classification maps on Houston task are shown in Fig 8.

3) Experiments on Shanghai–Hangzhou Task: As shown in Table VI, our classification approach exhibits a superiority of 5.49% compared to ED-DMM-UDA, which utilizes clustering techniques. In addition, it surpasses CDA by 5.73%, which considers class-conditional distribution, and outperforms CLDA by 2.73%, which considers the confidence of pseudo-labels. Our approach not only outperforms in terms of OA but also exhibits higher AA and kappa coefficient. Our method achieved an impressive AA of 92.91% and an exceptional kappa coefficient of 87.71%. The classification maps of different algorithms on Shanghai–Hangzhou are shown in Fig 9.

4) Experiments on Different Losses: In addition to conducting comparative experiments between different domain adaptation methods, we also conduct comparative experiments on DSL we proposed, domain adversarial loss (DAL), and MMD loss. Specifically, we replace the DSL in the SCLADA method to DAL and MMD for experiments. From Table VII, we can see that the use of our proposed DSL has achieved significant results on the Pavia dataset. The use of DSL increases by 0.99% and 0.87% on OA compared to using MMD and DAL, respectively. It has also improved in both

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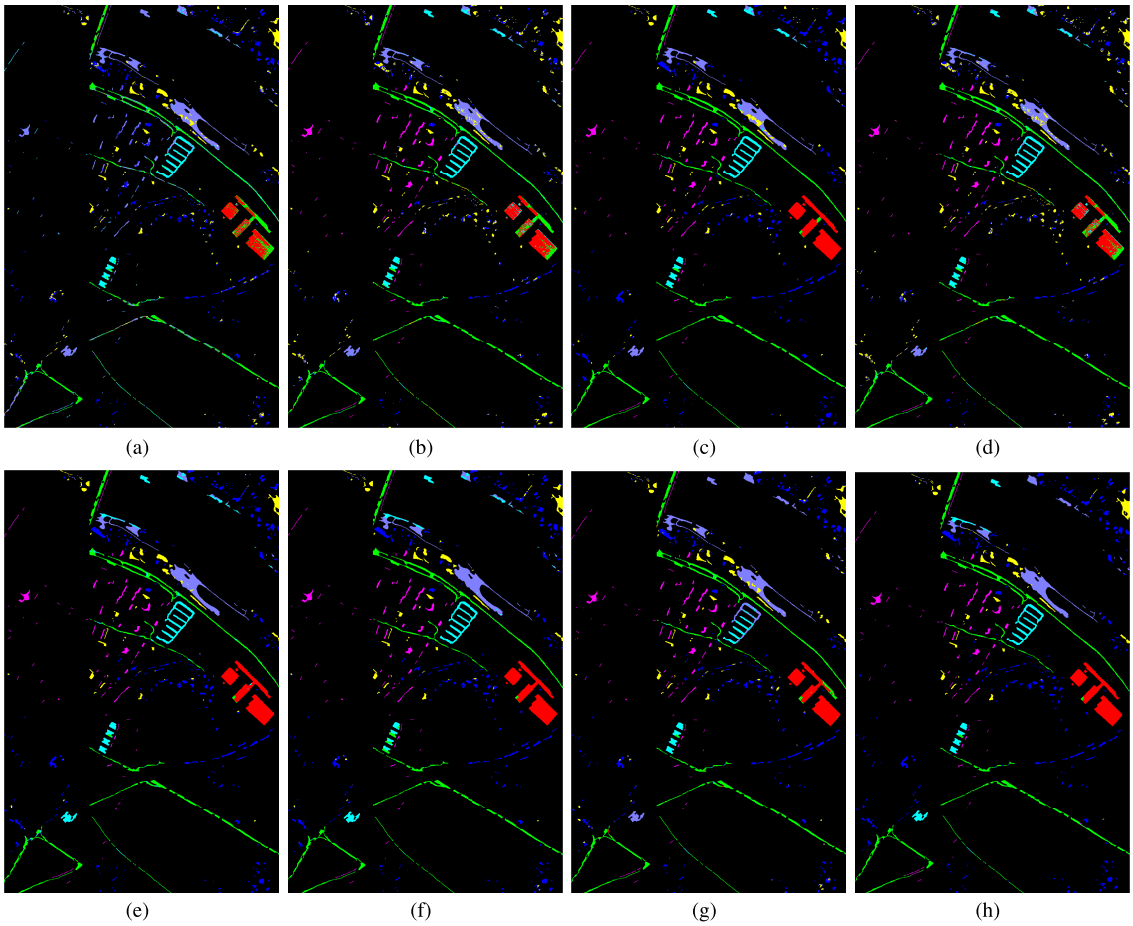


Fig. 7. Pavia task. (a) TSVM. (b) DAN. (c) DANN. (d) ED-DMM-UDA. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA.

TABLE VI
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE SHANGHAI–HANGZHOU TASK

Class	TSVM [16]	DAN [23]	DANN [27]	ED-DMM-UDA [29]	CDA [28]	CLDA [30]	TAADA [31]	SCLUDA
1	97.50	**99.84**	86.20	94.70	97.95	96.60	94.80	99.49
2	64.31	77.27	86.46	88.80	83.57	89.41	77.43	**92.77**
3	66.42	87.14	79.51	81.48	89.28	88.78	82.75	**90.22**
OA	82.43 ±1.97	83.20 ±1.83	84.37 ±2.32	87.42 ±6.30	87.18 ±0.84	90.18 ±0.98	81.31 ±1.70	**92.91 ±1.56**
AA	82.90 ±3.60	88.08 ±1.47	84.06 ±6.36	88.33 ±5.43	90.27 ±0.77	91.60 ±1.35	84.99 ±0.69	**94.16 ±1.62**
$\kappa \times 100$	69.28 ±3.83	72.52 ±2.83	73.06 ±4.20	78.12 ±6.91	78.39 ±1.35	83.05 ±1.56	68.73 ±2.20	**87.71 ±2.73**

AA and kappa. On the Houston dataset, although there are no significant difference in the classification results on OA, the use of DSL resulted in a significant improvement in AA for this method. On the Shanghai–Hangzhou dataset, using our proposed DSL, we still achieve the best results on all three indicators. In summary, the DSL we proposed can better achieve better performance.

D. Ablation Experiments

In order to further verify the effectiveness of SCL and fine-tuning-based confident learning, an ablation study was performed on each task, and Table VIII shows the results. In this case, Dl represents only executing domain similarity loss to reduce the discrepancy between domains, Dc denotes that solely the operation of domain adaptation for the class-conditional distribution is carried out, Cs and Ct represent SCL in the source domain and target domain, respectively, and F stands for fine-tuning based on confident learning. From Table VIII, it can be observed that performing both Dl and class-conditional alignment yields better results compared to using class-conditional alignment alone. It can be seen that using SCL in both domains is more effective than using it only in the target domain. In particular, the effect of fine-tuning based on confident learning is more significant. By fine-tuning, the accuracy rate of the algorithm is increased by 2.75%, 5.16%, and 2.80% on three tasks. Overall, each of our modules has played a corresponding role.

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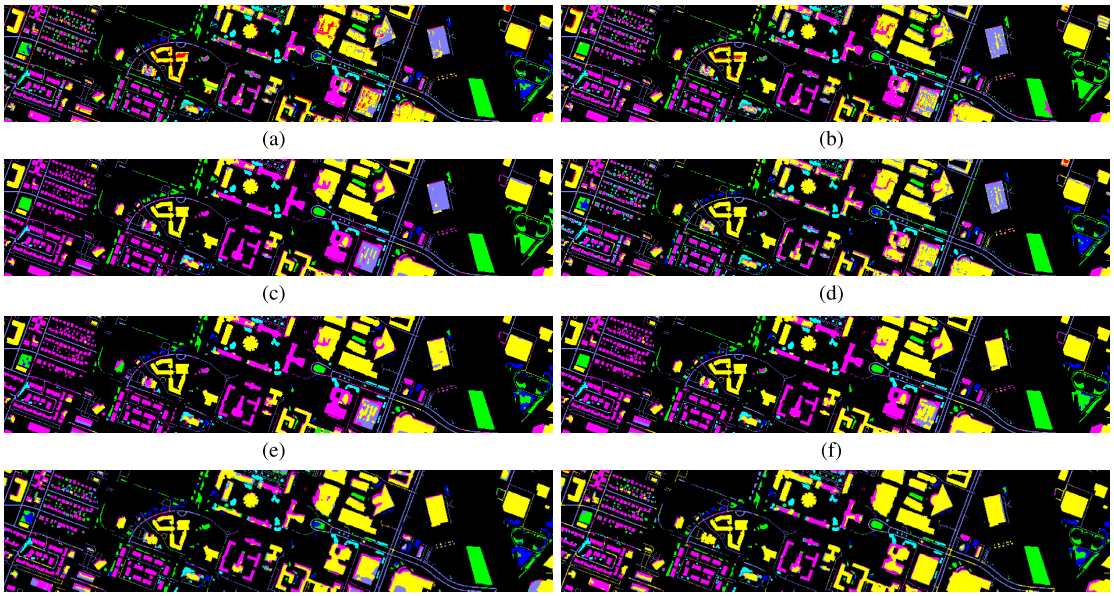


Fig. 8. Houston task. (a) TSVM. (b) DAN. (c) DANN. (d) ED-DMM-UDA. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA.

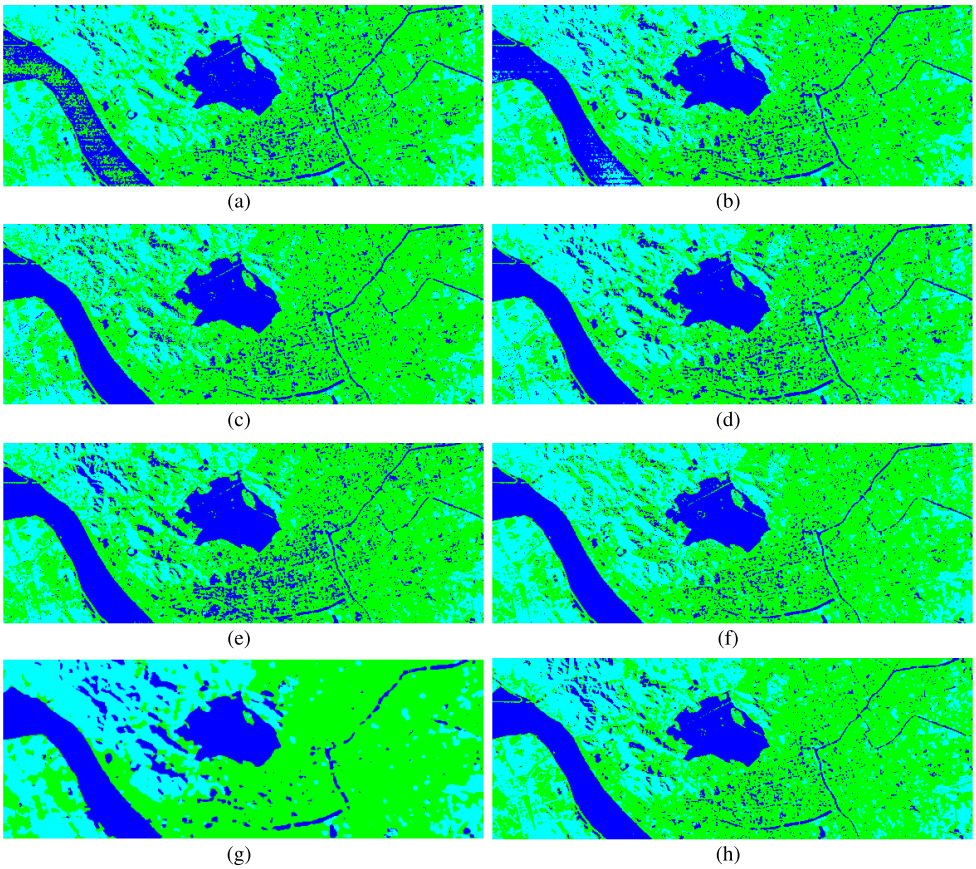


Fig. 9. Shanghai–Hangzhou task. (a) TSVM. (b) DAN. (c) DANN. (d) ED-DMM-UDA. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA.

E. Parameter Analysis

During our experiments, we identified six crucial parameters that required adjustment, namely, α, β, γ, patch_size, TRAIN_NUM, and learning rate. The α parameter governs the weighting of the LMMD loss, while the β parameter adjusts the weight of the DSL. In addition, the γ parameter controls the weighting of the cross-entropy loss used in fine-tuning-based CLM. We ensured that all other parameters remained constant while adjusting each parameter individually.

The α and β parameters determine the weighting of the domain adaptation loss function, enabling a quantitative assessment of the relative significance of each distribution. We tested eight different values for α, specifically 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.30, and 0.50. Fig. 10(a) shows the trend of classification accuracy with α changes. In the

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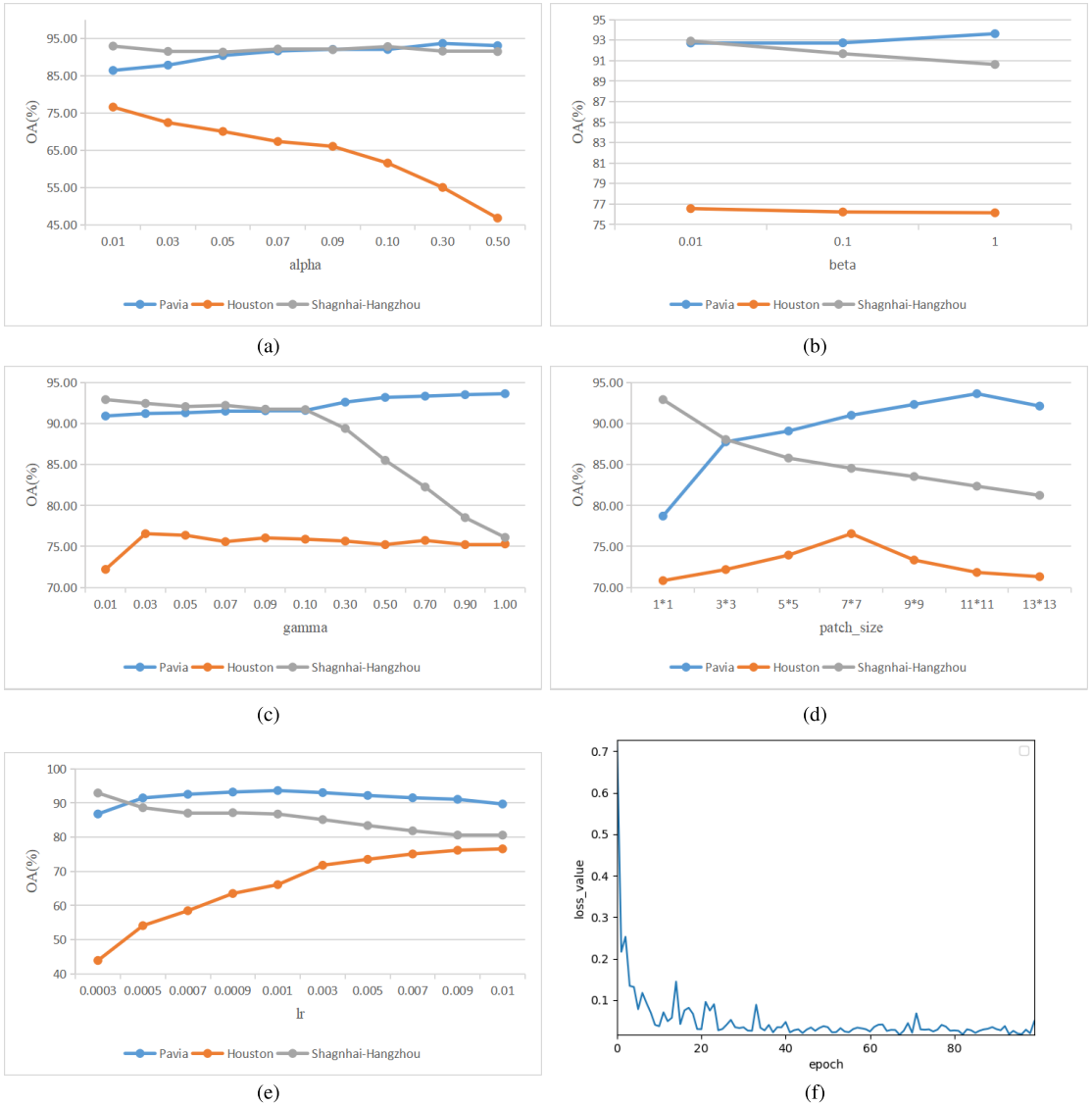


Fig. 10. Sensitivity analysis of parameters. (a) Parameter α. (b) Parameter β. (c) Parameter γ. (d) Parameter patch_size. (e) Parameter learning rate. (f) TRAIN_NUM.

TABLE VII
CLASSIFICATION RESULTS USING DIFFERENT LOSSES

Datasets	Loss	OA(%)	AA(%)	$\kappa \times 100$
Pavia	MMD	92.64±0.77	92.78±1.75	91.17±0.92
	DAL	92.76±1.22	93.01±0.95	91.32±1.43
	DSL	**93.63±0.34**	**94.06±0.88**	**92.36±0.40**
Houston	MMD	76.10±2.21	78.32±1.21	63.06±2.66
	DAL	76.44±1.10	77.64±1.95	63.24±1.13
	DSL	**76.56±1.83**	**79.20±1.24**	**63.98±2.19**
Shanghai-Hangzhou	MMD	92.22±3.02	93.24±2.90	86.43±5.31
	DAL	92.67±2.67	93.98±2.36	87.56±4.59
	DSL	**92.91±1.56**	**94.16±1.62**	**87.71±2.73**

TABLE VIII
ABLATION STUDY ON DIFFERENT TASKS

Dc	DL	Ct	Cs	F	Pavia Task	Houston Task	Shanghai-Hangzhou Task
✓					88.23±1.06	68.77±3.65	88.45±3.27
✓	✓				88.98±0.91	69.88±1.66	89.03±1.97
✓	✓	✓			89.99±1.09	70.45±1.90	90.04±1.71
✓	✓	✓	✓		90.88±0.69	71.40±1.81	90.11±1.65
✓	✓	✓	✓	✓	**93.63±0.34**	**76.56±1.83**	**92.91±1.56**

Pavia task, the accuracy showed an increasing trend with an increase in the α parameter, reaching its maximum value at 0.3. In the Houston task, the classification accuracy decreased with an increase in α. Thus, we set alpha to 0.01. In the Shanghai–Hangzhou task, a higher classification accuracy was achieved when α was set to 0.01. We tested only three different values for β, specifically 0.01, 0.1, and 1. The trend of classification accuracy in relation to β variations is shown in Fig. 10(b). Our experiments indicated that β had a minor impact on the classification results for the Houston task. Therefore, we set its value as 1. Based on these findings, we set the β value to 1 for the Pavia task and 0.01 for the Shanghai–Hangzhou task.

We tested 11 different values for the γ parameter, specifically 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.30, 0.50, 0.70, 0.90, and 1. Fig. 10(c) shows how the accuracy of classification

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TABLE IX
COMPUTATIONAL TIME (SECONDS), FLOPS, AND PARAMETERS ON EACH TASK WITH DIFFERENT METHODS

	Methods	TSVM	DAN	DANN	ED-DMM-UDA	CDA	CLDA	TAADA	SCLUDA
UP-PC	Training time	321.29	720.02	36.36	308.43	216.65	203.38	141.01	384.85
	Testting time	1.34	5.03	7.06	7.32	0.76	2.59	4.78	3.92
	FLOPS	-	27166983	16341960	3633402	20000000	3633983	50180201	80408762
	#params	-	23832839	13329224	181074	90260	182558	661880	1876017
Houston13-18	Training time	1847.56	818.73	36.58	1792.37	220.95	295.48	146.86	306.36
	Testting time	1.63	5.26	7.21	10.27	0.72	6.95	8.53	3.66
	FLOPS	-	25642887	14817864	3363402	20000000	3363983	7678089	15225218
	#params	-	23663495	13159880	170274	76760	171758	577640	1750305
Shanghai-Hangzhou	Training time	785.23	986.24	40.31	978.58	165.17	247.64	96.99	370.91
	Testting time	6.68	24.14	17.43	15.82	1.87	13.21	26.24	9.01
	FLOPS	-	24148483	13634756	201402	30000000	4113983	29503089	1453894
	#params	-	24125699	13628228	200274	113253	201758	811028	2097193

varies as γ is modified. In the Pavia task, the classification accuracy showed an increasing trend with an increase in γ, eventually stabilizing. Therefore, we set γ to 1 for this task. In the Houston task, the best result was achieved when γ was set to 0.03. On the Shanghai–Hangzhou task, the classification accuracy decreased with an increase of γ. Therefore, we set gamma to 0.01 for this task.

We tested seven different patch sizes, specifically 1×1, 3×3, 5×5, 7×7, 9×9, 11×11, and 13×13. The changes in classification accuracy with respect to variations in patch_size is shown in Fig. 10(d). In the Pavia task, the OA showed a gradual increase with an increase in patch_size, with the peak accuracy achieved at a patch_size of 11×11. In the Houston task, the OA initially increased and then decreased with an increase in the patch_size parameter, reaching the maximum value at a patch_size of 7×7. On the Shanghai–Hangzhou task, increasing patch_size led to a decrease in classification accuracy. Therefore, for this task, the patch_size parameter is set to 1×1.

For the learning rate, we tested ten different values, specifically 0.0003, 0.0005, 0.0007, 0.0009, 0.001, 0,003, 0.005, 0.007, 0.009, and 0.01. The trend of classification accuracy in relation to the learning rate variations is shown in Fig. 10(e). In the Pavia task, with the increase in the learning rate, the classification accuracy first increases and then decreases, reaching the maximum value at 0.001. Therefore, we set the learning rate to 0.001 on the Pavia task. In the Houston task, the classification accuracy increases with the increase of the learning rate and becomes stable when the value is 0.01, so we set the learning rate to 0.01. In the Shanghai–Hangzhou task, the classification accuracy decreases with the increase in the learning rate, so we set the learning rate to 0.0003.

For the TRAIN_NUM parameter, take the Houston task as an example, the loss curve without confident learning module is shown in Fig. 10(f). It can be seen that when the number of epochs reaches 40, the loss value tends to stabilize. Therefore, we set TRAIN_NUM to 40.

F. Computational Complexity

We have reported the training time, the testing time, the number of floating-point operations (FLOPS), and the number of parameters in Table IX for a comprehensive comparison of their computational complexity. All experiments are executed in a computer with a 3.7-GHz Intel Core I9-10900X CPU and a NVIDIA GeForce RTX 3090 GPU. Besides, the open-source deep learning framework Pytorch is used in the experiments. As shown in Table IX, the time complexity of our method is not significantly different from other comparison methods, and in this case, our approach has exhibited even more superior performance in classification tasks.

V. CONCLUSION

In this article, an SCLUDA framework has been proposed for cross-domain HSI classification. In SCLUDA, an SCL strategy is designed to enhance the separability of data itself within the source and target domains, which can improve the effectiveness of the alignment. In addition, we design a novel DSL based on OCC to handle the domain adaptation task, which can make the distributions of the source and target domains more similar, thereby reducing the discrepancy between domains. Moreover, in SCLUDA, we design a confident learning-based sample selection strategy to select high-confidence labeled samples from the target domain. By fine-tuning the domain adaptation model with these selected samples, the discriminability of the target domain model can be improved. Extensive experiments on three datasets show that the proposed SCLUDA outperforms other domain adaptation methods.

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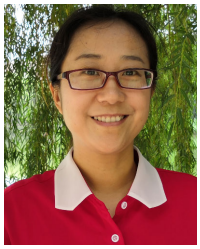
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